

The paradox of videotelephony — unconscious assumptions and undervalued skills

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What lies behind that querulous mixture of success and failure called videotelephony? Why has its popularity not matched its market potential, almost every time the idea has been tested? Developments in technology now mean it is time for a fresh look at video communications. Can the troubled past of the videophone be laid to rest and what can be done to make its future more productive and less turbulent? How do you untangle a knotted knot?

1. Introduction

Videotelephony is a paradox — succeeding in niche markets yet facing a future that is still unclear. If radio listening declined in popularity when television came along, why have videophones not replaced telephones? If people like to listen and talk with a telephone, seeing and being seen must surely have even greater appeal. The beguiling power of such expectations has endured to foster attractive yet implausible simulations (see Fig 1).



Fig 1 A 'rose-tinted' simulation of videotelephony.

All over the world, trials of videotelephony have given conflicting results, yet without sufficient clarity to force a fundamental re-think. There is little agreement about why, when or how pictures are most useful in telecom-

munications. Clearly, videotelephony is much harder to get right than conventional telephony [1, 2].

Yet videoconferencing, with carefully selected and high-quality equipment, is increasingly successful and attractive for meetings and group collaboration. This paper deals mainly with self-contained or desktop-computer videotelephones, meant for mass-market person-to-person communications. Videotelephones strongly exemplify the difficulties while excluding some special factors in the success of videoconferencing. The phrase video communication is used here generically, to include videoconferencing, telepresence, media space, virtual-world and other kinds of video systems.

The rapidly diminishing cost of video technology means that a fresh look at the design of video communication equipment is now due. Ironically, the strongest resurgence of interest comes from outside the video and telecommunications industries. The computer industry has brought a different design philosophy, where imaginative dreaming and multi-minded synthesis are given greater weight [3, 4]. The success of this approach, in finding new markets and applications, implies that users have probably not had a big enough say in development.

How did this come to be? The authors argue that videotelephony development has been distorted by the inappropriate consequences of unconscious assumptions. Three kinds of assumption seem to have directed, some would say bedevilled, four decades of international effort in videotelephony. These assumptions are about:

- classification — the type of innovation involved,
- expectation — of purpose and likely performance,

- terminology — the words, names and concepts used.

These assumptions are self-reinforcing and entrenched. They seem to have inhibited broader study of where, in collaborative communication, the value of a picture lies. What should the picture express and to what degree? Four kinds of skill seem to have been undervalued in conventional studies of video communication. These skills concern:

- physical and psycho-physical optics — in cameras and displays,
- pictorial culture — the artistic aspects of picture control and composition,
- expression — how meaning and content are represented by the picture,
- projection — how the picture conveys psychological issues between users.

This paper debates these unconscious assumptions and undervalued skills, exploring their implications for the future. Some options for new explorations in video communication are then presented. The story starts with something that seems innocuous yet has been fundamentally misleading.

2. Classification

2.1 *Old or new?*

What kind of thing is videotelephony, an old thing or a new thing? Telephones and televisions are well-established devices, whose use is understood, so they count as old things. When AT&T launched the PicturePhoneTM service in the early 1970s [5], it was seen as an obvious next step in telephony. Videotelephony was therefore seen as a new use for old things. The authors think this classification is the root cause of problematic development in videotelephony. Video communication should properly be seen as an old use, namely, human communications, for a new kind of thing created by the exchanging of electronic pictures.

2.2 *Exploiting and exploring*

The exploitation of existing ideas is quite unlike the exploration of new ones. The social, engineering and commercial aspects of the design and development process change completely. For example, an electric motor for a new application can be designed and refined by the manufacturer, with no user trials of any kind. The motor has a new use but is an old thing, with design and performance requirements that are easy to specify. In the exploitation of old things, suppliers lead users to new applications and devices can be designed before they are made.

Alternatively, imagine that something called a 'polytron' has been invented. Crude models are produced for demonstration, followed by early kinds of polytron sold to a small market of enthusiasts. If the idea proves popular, the market grows despite the fact that no-one really knows how to use or build polytrons well. Indeed, there is much debate about when and where they should be used. Many years later, when their social and cultural impact is clear, the optimum engineering design for polytrons is established. When something new appears, users lead suppliers to new understanding and devices are made before they can be designed.

2.3 *A lucky accident*

The telephone was a new thing. No reliable way to transmit the human voice over great distances existed before. Early telephone users, with crude and inefficient types of telephone, decided how, why, when and where telephony should be used. By accident more than design, these crude telephones were sufficient to create new kinds of social behaviour and working practice — such as cancelling meetings at the last moment or relying on distant experts to solve immediate problems. Detailed scientific understanding of good design was achieved many decades after the popular use of telephones had defined their social purpose.

2.4 *Classifying video*

Extending the range of human speech augments human ability; but the benefit of extending the range of sight, for ordinary conversation, seems superficially less compelling. It has often been described as an added luxury for telephony. When a new thing is employed for an old use, that use can be revealed as less well understood than had been thought.

The authors believe electronic pictures can transmit information that is just as relevant as the information of speech; but this information is different and more varied. It can be circumstantial and abstract as well as factual and concrete; it can support meaning as illustration or modify it as context; it can be involuntary as well as carefully selected and it can sometimes be less than beneficial. Videotelephony is, in some ways, telephony with video yet in other ways, oddly, it is not. The use of video pictures for human telecommunications creates a broad and confusing mixture of old and new kinds of purpose — a mixture which only recently has begun to be explored [6] (see Fig 2).

But worse than this mis-classification of videotelephony, as telephone with television, has been the resultant creation of a raft of unrealistic expectations, in cultural, social, technical and commercial terms.



Fig 2 Exploring telepresence.

3. Expectations

3.1 Fair comparison

The telegram was never berated for lacking the subtle expression of a letter. In early demonstrations of telephony, poor sound quality was compared with the inconvenience of travel. The international measure of telephony performance, the mean opinion scale rating, is based on comparison with a high-quality wideband connection not face-to-face conversation. Yet videotelephony is conventionally measured through comparison with face-to-face contact — and many studies have found that videotelephony is inferior, even with the highest possible video quality. The seductive power of a picture is so strong that unrealistic or impossible expectations have arisen.

3.2 Aiming too high

Some studies of video communication have assumed that the picture will have an impact like the best of the cinema and television, where viewers can feel a sense of involvement in a distant place. The terms ‘virtual reality’ and ‘natural presence’ are sometimes used to mean an experience almost indistinguishable from physical presence. The authors think this interpretation is unrealistic and unhelpful in videotelephony. It tends to raise expectations and to gloss over important and complicated problems. The entertainment industry seeks bigger and better sensations, but reality is never on the bill.

The term ‘virtual reality’ is widely associated with immersive environments, that use helmets, headphones, force-feedback gloves and motion platforms. It would be useful to use the term ‘natural presence’ to mean a less physically immersive but psychologically captivating kind of environment, where users can feel a sense of involvement with distant people or events. It is possible that some kinds

of video communication might be ‘better than being there’ [7] for a particular purpose, without trying to be ‘the same as being there’.

3.3 Understanding purpose

Video communication should be seen as a thing in its own right, not as a poorer kind of reality. The question is not whether videotelephony can emulate face-to-face meeting, but what pictures can add to human interactions. Richer and more effective forms of collaboration can occur with sensations that are smaller than real life yet much larger than ordinary telephony.

A closer approach to reality is not always an advantage in telecommunications. Ordinary telephony allows users to be mentally captive yet physically free. Face-to-face meetings have other drawbacks than the need for travel. For example, it is rude to manifestly allow attention to wander if the discussion becomes uninteresting. But users in an audioconference can divert part of their attention to minor business without embarrassment.

Deeper consideration of purpose in videotelephony means identifying the positive aspects of face-to-face conversation and of travel, that are useful to emulate. This may lead to lesser-explored areas of social psychology rather than well-explored areas of high-quality video technology. These aspects can be strongly circumstantial, dependent on purpose and task — if there is a task.

3.4 Interacting and adapting

In television and the cinema, viewing is passive and non-interactive. Transmission is in one direction, with pictures that are deliberately controlled and selected. The distinction between performer and observer is clear. Most applications of videotelephony involve highly interactive viewing, where transmission is in both directions. The pictures are not actively selected or composed by anyone and the observers are also the performers. Being seen can have a marked impact on the sensation of seeing¹. This is why comparison with cinema and television has led to unsuitable performance metrics.

An important consequence of interactivity is that, with enough motivation, users will adapt to limitations in technology. In the developed world, the contrasting sounds of real people and ‘telephone people’ become reconciled in childhood. The motive for such adaptation is powerful — to match adult skills in communicating — so the process becomes part of growing up and is seldom noticed. Will users adapt to the characteristics of videotelephony? What might be sufficient motive to learn to cope with more numerous and variable limitations?

¹ Viewer and viewee issues are discussed in section 8.

3.5 Success at last

Videotelephony is quite different from watching television and very different from conventional telephony. But these are not good reasons to avoid videotelephony. To borrow the style of Abraham Lincoln — ‘You can berate some videophones all of the time; and you can berate all videophones some of the time; but you can’t berate all of the videophones all of the time’.

Those early expectations, that videophones would replace ordinary telephones, have not been fulfilled. Instead, experience has shown that videotelephony can enhance communication in a new way, without supplanting face-to-face contact. Ordinary telephony has proved superior for some purposes, yet videotelephony has found use outside the boundaries of telephony [2].

Videoconferences are typically more formal and organised than telephone calls, often with a chairperson in charge of the discussion. Many companies, large and small, have found that videoconferencing can be a very effective way of co-ordinating geographically separated teams. There is also abundant evidence for the commercial success of video communications in a growing range of unconventional applications in education, training and medicine (see Fig 3) [8–10].



Fig 3 Frank Dobson watches a foetal scan at the Women and Children’s Centre in Deptford Market, with a video link to experts in Guy’s and St. Thomas’s hospitals [© David Allen/WayMark].

Many of these expanding applications do not fit the conventional expectations of the historical ‘telephone with television’ model. Forming appropriate expectations for video communication means developing a broader and deeper understanding of the task not the technology. This implies a need to re-examine, and expand, the terminology that describes visual telecommunications.

4. Terminology

4.1 Words and names

Simple words can create complex misunderstandings. The word ‘videotelephony’ suggests two of the most successful inventions of the century. Someone whose parents are called Einstein and Thatcher is probably going to have a hard time as a scientist or as a politician. Unconscious assumptions are, of course, the hardest to detect and change. When things are assumed and unstated, generalisations can turn right into wrong.

The words ‘video’ and ‘communication’ seem to have been used, in academic and commercial study of videotelephony, as if they meant well-developed and tightly-specified things like televisions and telephones. But they do not. In telephony, the terms ‘transmitter’ and ‘receiver’ have long been used to mean the well-specified and particularly-located microphone and earphone in a handset. In videotelephony, generic terms like ‘camera’ and ‘display’ are still universally used, despite precise specification of coding and networking standards. Should videotelephony use special names for particular kinds of camera — like ‘portrayer’ or ‘landscaper’?

4.2 Conflicting results

In videotelephony, decades of world-wide academic study and practical trials have given confusing and contradictory results [2]. Different investigators have found video useful, partially useful and useless as an addition to telephony. It seemingly can add social but not emotional value or add emotional but not social value, in a way that is much disliked while being generally popular.

Contradictory answers imply that the questions were not well specified, or that unsuitable measurements were made. Some studies may not have been comparing like with like, suggesting that knowledge of how to use video technology is immature. The authors believe that inadequate or misapplied terminology is one important reason for these problems. The words we conventionally use do not succinctly define what these systems do, nor how they should be used. The terms ‘telepresence’ and ‘media space’ are two attempts to find more suitable words.

4.3 In human terms

Video and audio signals are sometimes treated like different members of the same species, differing only in superficial matters of bandwidth and signal-to-noise ratio. This is a gross distortion of reality. Audio and video signals have very different technical and human characteristics, in their conception, production and perception. Table 1 shows the properties of audio and video signals when created and interpreted, from a human perspective.

Table 1 Human properties of audio and video signals.

	Transmission	Reception
Audio	Strongly cognitive Controllable	Cognitive Uncontrollable
Video	Mainly affective Circumstantial Mainly uncontrollable	Affective Cognitive Controllable

Speaking is cognitive and easy to control — calm phrases can accompany inner turmoil. It is much harder to consciously alter how we look — emotion tends to form our appearance and behaviour. Circumstances can lead us to choose our words with care but can dictate our appearance or behaviour without choice. Listening is mainly cognitive, with emotional overtones, while looking can be cognitive, emotional or somewhere in between. It is impossible not to hear what is audible, yet it is easy to avoid seeing what is visible.

This marked asymmetry of human properties implies that some of the technical skills of audio communication may not be useful in a video context, nor may the

terminology. The dominance of the cinema, television and telephony, in supplying ready-made models and terminology, has diminished the recognition of this asymmetry and its impact in videotelephony.

5. Viewpoints

Unconscious assumptions in the history of videotelephony are summarised in the viewpoint of Fig 4, while Fig 5 shows a different viewpoint where human values, behaviours and opinions are the important and difficult questions. This shift of emphasis mirrors the different approach to technology development suggested by Lewis [11].

Remarkably, both these viewpoints are found in a book written twenty-five years ago, by Edward Dickson of Cornell University [12]. In the early chapters, Dickson makes the now conventional unconscious assumptions about videotelephony, focusing on the technical problems of transmission. Yet the later chapters contain a clear and wide-ranging summary of the subjective and social issues, focusing on user requirements and opinions. A broad and imaginative range of ideas for non-telephony applications is

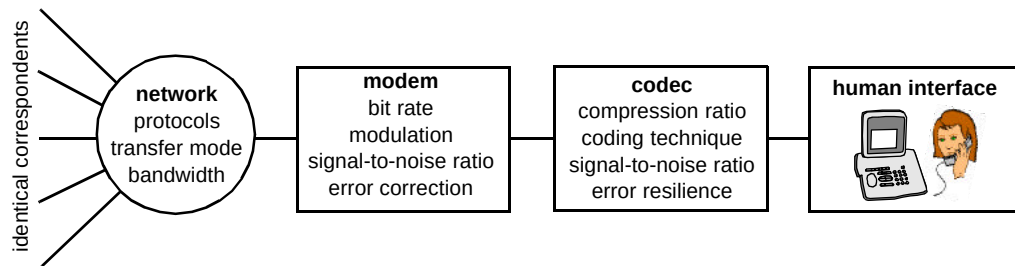


Fig 4 The technocentric view of videotelephony.

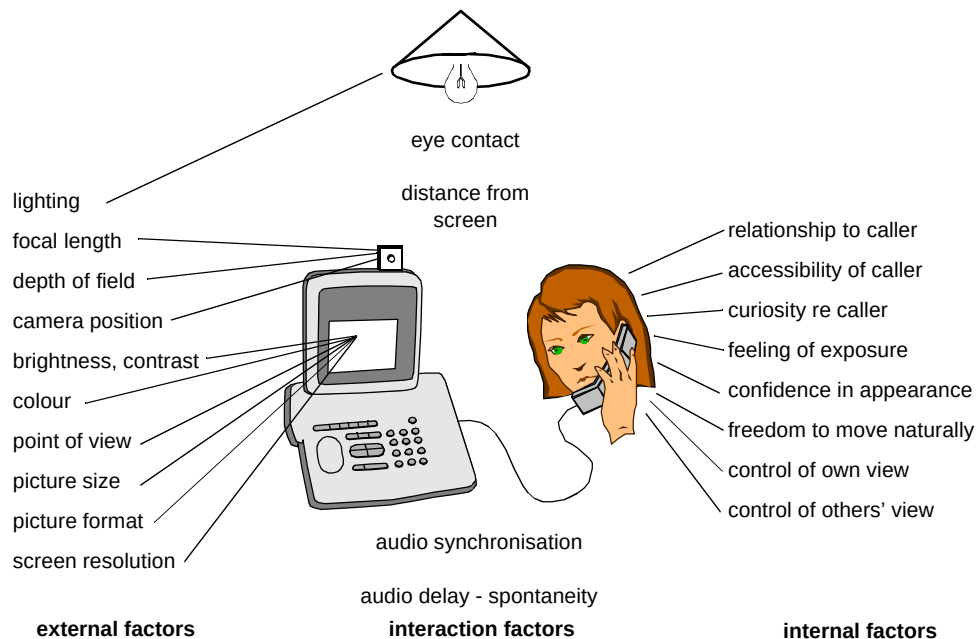


Fig 5 A human-centric and motive-driven view of videotelephony.

also given. All the important areas for research and development are recognised in this book, yet the seeds of misadventure are obvious.

Attention has been diverted from four types of skill or knowledge about video communications, that are examined next.

6. Physical skills

6.1 *New parameters*

Physical parameters in ordinary telephony, such as loudness, frequency response, noise, reverberation and echo are complex and interdependent. After decades of international study, these parameters are widely understood and recognised. Capturing and displaying pictures involves a greater number of parameters, which are well understood though less widely recognised. Many of these parameters have no exact analogue in ordinary telephony. Technical development in video communications has focused on signals and coding, the last of the parameters below, to the marked detriment of other issues.

6.2 *Optical perspectives*

The location of the camera — the viewpoint — strongly affects both the technical and subjective qualities of the picture, radically changing the perspective the viewer sees. The focal length of the lens, with the dimensions of the image sensor, fix the camera's angle-of-view. The camera-to-subject distance then determines the proportion of the picture occupied by the subject, from a close view of the face to a full view of the surroundings. There is no single optimum proportion, because facial expression and visibility of body language and the environment are all important, in different ways at different times. A fixed compromise is usually provided, though some cameras allow adjustment. The aperture of the lens alters the depth of the field of focus. Small apertures are commonly used, to show as much as possible in focus, yet blurred backgrounds can concentrate attention on to the subject.

6.3 *Lighting*

Every good photographer knows that lighting is vital in successful portraiture. The nature and location of the lights changes contrast, colour and the fall of shadows in a pronounced way [13]. Bad lighting can have a very deleterious subjective effect, creating images of people that are pale and featureless or stark and haggard. Little attention has been paid to lighting in videotelephony, where the technical ability to capture noise-free pictures under ordinary room lighting has been seen as beneficial. Attention to lighting is usually part of the installation of videoconferencing equipment, where capturing accurate or flattering pictures is recognised as important.

6.4 *Under control?*

Adjustments for contrast, brightness and saturation are a familiar part of every television and desktop-computer display, to compensate for local display conditions. Yet few desktop computers provide control of a parameter called gamma², despite the fact that different manufacturers use different values and none of these is compatible with gamma in television. As a result, pictures exchanged between different computers, with different gamma values, can appear uniformly too thin or too gloomy in appearance. Should the viewer make all the adjustments, the viewee or neither?

6.5 *Size and resolution*

In television, large displays imply status and the common assumption is that larger displays are always better. For video communication, small displays can have advantages. They tend to conceal picture impairments and can be more convenient to accommodate. In video communications, display size should be fit for purpose and depend on viewing distance. Resolution is a function of camera, transmission channel and display factors. A common assumption is that higher resolution pictures are always better, but in video communication resolution should be sufficient for purpose.

Knowledge of the human visual system and of psycho-optics is important in determining what size and resolution of picture matches a given purpose. Large and high-resolution pictures of faces can be unflattering or have unintended psychological side-effects. A small, highly-resolved picture contains the same information as a large and poorly-resolved one, yet subjectively the smaller picture appears to be of higher quality.

Display size and resolution are often quoted in ways that seem exact but are actually vague. The intended viewing distance should always be quoted, if display size is given in linear units or when resolution is measured in pixels. A better approach, which integrates the effects of resolution, display size and viewing distance, is to quote both resolution and image size in units of angle — measured at the viewer's eye. This allows a simple estimate of system magnification, given the likely distance between the users if they were face-to-face.

6.6 *Coding complexity*

Coding for bit-rate reduction has been seen as the crucial economic issue in video communication for decades [14]. Other aspects of video communications, including user-related factors, seemed simple by comparison with the

² Gamma controls the compensation for nonlinearity in the camera and display. Incorrect setting distorts the distribution of light and dark shades.

difficulty of efficient coding. Early video codecs were huge and expensive pieces of hardware, even though the bit-rate reduction was modest [15].

Enormous improvements have resulted from concentrated effort, world-wide, in image coding [16]. Descriptions and analyses of run-length coding, discrete cosine transforms, predictive coding, vector quantisation, fractal coding and numerous other techniques have clogged the technical journals for decades. Some of this work has been enshrined in international standards; the best known of which is the JPEG still-picture standard [17], which typically offers a thirty-fold reduction in data size without objectionable impairment.

Moving pictures present a greater challenge, because of the movement and the different video systems in use. Coding schemes have evolved, culminating in the successful group of standards that includes the MPEG1, MPEG2, H.261 [18–20] and the still-evolving MPEG4 and H.263 techniques [21].

6.7 Coding acceptability

Video-coding algorithms have become sophisticated and the computing power they need is now affordable. Yet problems remain. Imagine striving to invent the aeroplane and fly over the ocean, only to face a fifty-mile jungle walk from the airport to your destination. The distance left to travel is small, yet the task is daunting.

Removing redundancy from pictures has proved to be a straightforward mathematical task. Understanding human reactions to minor picture imperfections is still a difficult problem — people relate to images in complex and very subtle ways [13, 22]. Coding quality is defined by the subjective acceptability of the distortion artefacts introduced by the coding. The conventional measures of objective distortion, such as rate-distortion and root-mean-square error, are poorly correlated with these human opinions. The attraction of fractal methods for picture coding may be less connected with their inherent efficiency and more with visual plausibility of fractal coding distortion.

6.8 Signalling a difference

Despite decades of technical focus, picture signals are still poorly understood. Finding a closer connection between the human and technical aspects of pictures means improved understanding of both subjective and objective issues. For example, pictures are commonly said to need much more transmission bandwidth than speech, but the real difference is far deeper. Speech signals are one-dimensional (voltage) scalar functions of time while colour picture signals are three-dimensional (red-green-blue) matrix functions of time. The order of the matrix is the

number of spatial samples (pixels) in the picture — 307 200 for a VGA-size image. Mathematically and cognitively, information in moving pictures is dispersed over a much larger and more complex signal space than is the information of speech. This has important consequences not only for processing and coding pictures but also for attempts to link human opinions to technical features.

A single time-sample of a speech signal carries no information whatever. Patterns of evolution over time carry the information in speech. A single time-sample of a moving picture carries a lot of information — it is a still picture. Yet a composite PAL video signal is, like speech, a one-dimensional scalar function of time. It is produced from a picture signal by serialisation, in scan format, of the spatial-samples of each picture matrix. Three scalars are then combined into one by sub-sampling and quadrature sub-carrier modulation. Video signals are therefore a subtle weave of time and spatial sampling, disguising the true nature of the picture signal.

Video signals are strongly periodic, but this is almost wholly an artefact of the serialisation and modulation that turns three large matrices into one scalar. Pictures of fences and fabrics occasionally have spatial periodicity, but most pictures are not periodic.

7. Pictorial skills

With pictures, both eager reading and creation are normal parts of childhood; but words and numbers soon displace pictures in education, all the way to adulthood for most people. In western culture, pictures are mainly for viewing and the creation of pictures is seen as needing specially gifted or artistic people. In previous centuries, things were different; and in other cultures they still are. Creating effective pictorial composition, perspective, balance, tension, drama or relaxation is the stock-in-trade of photographers and cinematographers. Television pictures can show good pictorial skills too, though visual impact and operational convenience usually matter more.

This cultural bias has meant that pictorial skills have been neglected in videotelephony [1]. In the earliest types of videophone, the camera and display were often fixed and the equipment was in charge of the user's behaviour and location [23]. This usually resulted in 'square-on' shots of faces in tight close-up, like a passport photograph, as in Fig 6. Few amateur photographers take pictures like that, because backgrounds and bodies usually feature too. Cynics have said that artistry in videotelephony used to be measured by how far the subject's nose was from the centre of the picture. When pixels are scarce, artistry might be a luxury, yet relatively few pixels are needed to recognise a face.

Fortunately, pictorial skills are growing in video communications, as affordable cameras become adjustable and more mobile. But some fundamental issues of equipment design, about the degree and disposition of the users' control of picture composition and content, are poorly explored. As well as the technical and physical issues, that were discussed in section 6, there are psychological and psycho-physical issues in communicating through an exchange of pictures.

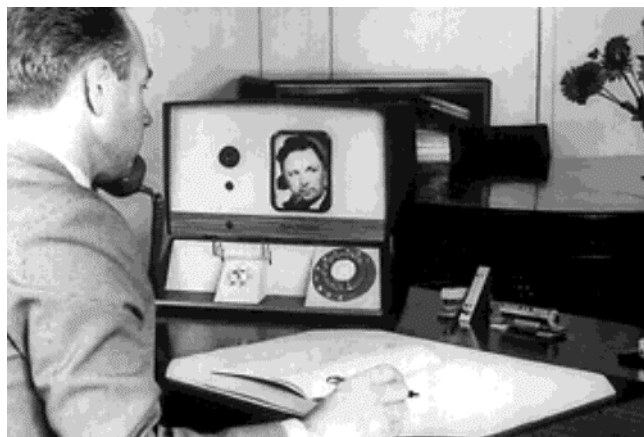


Fig 6 An early videophone prototype, using a close-up view of the face.

8. Features in expression

8.1 What is expression?

The main purpose of the picture in video communications is to give expression in support of communication. Expression is meant to be a generic term, describing different things at different times and in a variety of ways, that depend on the intentions of the users. Pictures in videotelephony can express facial detail, body language, the circumstantial conditions or illustrate objects under discussion³. Expression is mostly an internal property of the picture, i.e. something which an external observer might discern from the picture alone. Expression is the simplest answer to the question: 'What is the picture for?'

Expression in videotelephony is often asymmetric, because control of how expression is sent (being seen) can conflict with control over how expression is received (seeing). This has no direct analogue in telephony, where control over talking conditions has little connection with listening conditions and vice versa.

³ 'Video-as-illustration' or 'Video-as-object' are proposed as better phrases than Nardi's 'Video-as-data' [2]. The word data is already widely used in telecommunications and computing to mean bit streams and files. It suggests impersonal attributes that can be misleading in the context of video communications.

8.2 Receiving expression

The expressive effect received, by someone looking at a picture of another person, is a complex subjective function of the physical and pictorial parameters of the subject and the background. Additionally, the local environment in which the picture is displayed can change the impact received. There are also indirect issues, such as the psychological effect of distortion to eye contact that occurs when the camera and display are not collocated. The edges of the reproduced picture also limit the viewer's ability to focus attention on different parts of the distant location. Viewers could be given control of the look direction and angle of view of the distant camera, but the viewee might find this disturbing or offensive. Received expression can be optimised by negotiation between the parties, about the comfortable setting of a range of parameters. This process is seldom part of practical videotelephony and ordinary telephony has almost no need for mutual adjustments.

8.3 Sending expression

Controlling the intended expression is straightforward when speaking, but most people are less skilled — and less comfortable — in controlling how they appear in front of a camera. It is harder to select, reveal, conceal and prioritise what we visually express. Aspects of our body language and surroundings can be beyond voluntary control [24]. Part of this lack of skill results from unfamiliarity with the situation. Much of the difficulty, however, is fundamental, in that there is no easy way to put ourselves in the eye, or mind, of the beholder. A self-view picture can help, though this can also be distracting.

Being interrupted by a conventional telephone call means suffering a limited amount of intrusion, even if accepting the call is inconvenient. An unexpected videotelephone call involves a much greater degree of potential intrusion, if we are not dressed appropriately or are in surroundings that are untidy or unattractive. This can create the stressful or menacing feeling of being spied upon, although establishing new social norms for videotelephony could reduce the impact of this problem (see suggestions in section 10.3).

Users of videotelephones typically show much greater willingness to see than to be seen. This asymmetry between seeing and being seen is the opposite of the asymmetry between listening and talking. Most people find sending speech less stressful than receiving it, yet trying to send visual expression is much more stressful than receiving it, in emotional and cognitive respects.

Giving the viewer increased control of received expression directly reduces the viewee's control over how and what expression is sent. This is a human problem not a technical one, though solutions are possible through

negotiated agreement or compromise on technical factors. This problem is entirely ignored in conventional types of videophone [5].

9. Angles on projection

9.1 What is projection?

The word projection means throwing or extending, which in video communications implies things connected with a sense of distant influence or surrogate presence, as provided by the picture. Projection can also mean mapping of mental states on to objects or other people. Projection is used as a generic term for how users exploit the picture, to convey influence or exchange psychological or emotional attitudes. Projection is mostly an external property of the picture, i.e. something which is visible in the picture only in the context of the communication system as a whole. Projection is an alternative, more subtle and complex answer to the question: 'What is the picture for?'

Facial expressions, gestures and body language carry emotional or affective information about someone's state of mind or psychological outlook. Complex and mutually interrelated patterns of this behaviour emerge in conversations, varying in synchronism with the meaning and intonation of what is being heard and said (see Fig 7). In person-to-person video communication, the two pictures and the two speech signals collectively represent the transmission, or projection, of this behaviour as an ensemble.

Participants in a conversation automatically acquire a mental model of how they are 'coming across' in influencing or connecting with others. Video communications systems with good projection allow users to 'come

across' without special effort, enacting and reacting to affective information in a natural way.

Issues about projection are common to video communication with live pictures, live pictures with computer-generated backgrounds and with wholly synthetic pictures of avatars. Because projection involves shared motivations, conflicts in the control of sending and receiving conditions are infrequent in person-to-person calls, though they can arise in group communications.

9.2 Relative behaviour

Projection is an undervalued and under-explored aspect of videotelephony, strongly influenced by the intentions of the users and the nature of their relationship. Circumstantial factors can inhibit or stimulate affective interactivity and therefore change the need for good projection. Projection in video communication always requires good expression, but the converse is not true if affective information plays little part in the call.

Typical factors that interact to decide whether good projection is important in video communication include the physical distance of separation between users, the emotional distance between them, when they last met — if at all — and the relevance of appearance, body language and circumstances to that call. For example, changing an appointment with a colleague is usually routine, with little need to show or monitor emotion; but talking to a loved one a long way away, who is dealing with a stressful problem, is likely to create a strong wish to exchange live pictures. Equally, a call from a stranger might well create curiosity about their appearance, as an ordinary part of social interaction, though a still picture might suffice.

These factors are illustrated in Fig 8, where the axes show the strength of different aspects of the situation. Note

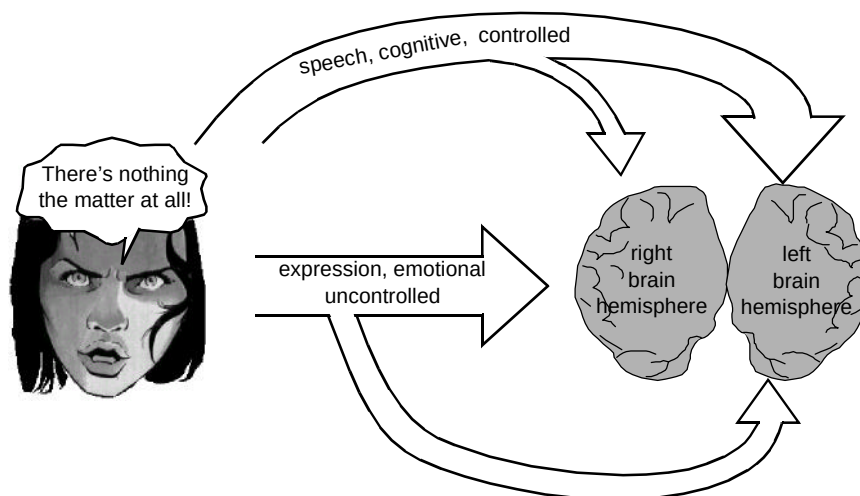


Fig 7 Flow of information and affect, to the left and right halves of the brain.

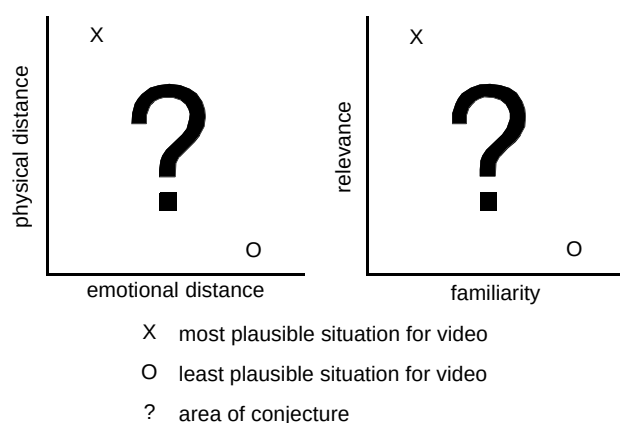


Fig 8 Physical and emotional distance, or relevance and familiarity, alter the likely appeal of videotelephony.

how much of Fig 8 is uncertain ground, because the boundaries between plausible and implausible need for projection are still unsure. Academic investigation of these factors has commonly focused on specific and pre-determined kinds of task. Less formal user experimentation might foster feedback which, though anecdotal and imprecise, would cover a wider variety of purpose. Such exploratory or attitude-forming kinds of user trial have frequently been undervalued in videotelephony.

9.3 *Your place or mine?*

What kind of projection happens in ordinary telephony? Where, mentally, do telephone calls take place? Does the telephone bring the distant user 'here' or is the local user taken 'there'? The universal appeal of telephony, and much of its power, may lie in its ability to take both users to an imaginary 'somewhere else'. That place seems to automatically acquire a mix of the characteristics of both 'here' and 'there', as territory that is more mutual than neutral. In football terms, the match is played at a friendly third-party ground.

The addition of pictures, and the sense of reality they bring, might change this situation. When a solo user joins a distant group by a video link, that user is probably going to feel taken 'there' by the link. The solo user is playing an away match.

What happens in person-to-person videotelephony? Do the two users change places when each is taken 'there', or do they seem to meet in the middle? The authors suspect that both users feel the other has been taken 'here'. Both teams are playing a home match — which is inherently contradictory. This could create kinds of conversational behaviour that are mutually incompatible. This line of reasoning implies that videotelephony genuinely is a paradox, because adding real pictures diminishes any mutually compatible sense of imagined meeting. If so, pictures that insert live images of people into synthetic backgrounds might increase the appeal of videotelephony.

10. Future development

Fresh study of videotelephony, as a new device for an old purpose, should focus on the less tangible, user-related issues of content and motive. Such studies would help transcend the limitations of the traditional, technically oriented approach. The authors also propose the study of simple but standardised ways for users to control call progress.

Initially, new studies should be open-ended and flexible, with more emphasis on encouraging user playfulness than on accuracy of measurement. Studies could be longitudinal in scope or use unconventional assessment methods to explore neglected aspects of video telecommunications — aspects which are questions about expression and projection.

10.1 *Expression*

What should the picture express? What should it be a picture of and why? How suitable is the expression which is achieved? These questions are fundamental to making video communication more attractive and productive.

The answers depend, for example, on knowledge about the nature of the relationship between the users, the purpose of the call and the context in which the call is made. These issues are likely to be quite variable, between calls and even within a single call — which implies that a range of picture types should be available. The technical and artistic aspects of picture expression include:

- the angle-of-view and location of the camera,
- the balance of visibility between facial detail and body language,
- the apparent size of the reproduced picture,
- the integration of the picture with sound, and the sound quality,
- the style and composition of the picture,
- the control, by the user, of what is seen and how it is portrayed,
- the quality of the reproduced picture, and the need for motion.

10.2 *Projection*

How does the picture convey, between users, the psychological or emotional factors in the communication? Is good projection more important for the viewer or the viewee and why? How useful is the kind of projection that is achieved? These questions are fundamental to the

psychological role of the picture, and to its relevance to the users.

Answers to these questions depend, for example, on knowledge about the kind of communication between the users, their physical and psychological circumstances, the extent of their acquaintance and the relationship between them. Furthermore, the requirements for projection may be quite different for person-to-person, person-to-group or group-to-group conversation. The technical and artistic aspects that influence projection through pictures include:

- the field of view and location of the camera,
- the degree of formality or informality implied by the picture,
- the suitability of the context of the picture, as taken and displayed,
- the size and location of the reproduced picture,
- the balance of visibility between facial detail and body language,
- the nature, direction and colour fidelity of the lighting,
- the control, by the user, of what is seen and how it is portrayed,
- the degree of synchronisation of picture and sound,
- the amount of movement distortion due to bit-rate compression.

Some features affect both expression and projection, though not necessarily in the same way or to the same extent.

10.3 *Call progress*

To strengthen the quality of expression and projection in video communications, the user should both feel and be in control of picture presentation. This means more than controlling the composition, selection and display of pictures, even though these issues are fundamental. User control in videotelephony starts with a simple question: 'How do you answer a videophone?' It is surprising, and revealing, that no socially acceptable procedure has yet been developed for this small but psychologically very important task. Options for answering videophone calls include:

- an initial answer, by convention, with a still-frame or pre-recorded movie,
- the option of switching to live video, as a mark of privilege or of welcome,

- the option of a deliberately degraded (small) live overview, for confidentiality,
- choosing to remain in still-frame mode should be acceptable, as a right to privacy,
- the control of viewpoint, in what to show and when (possibly negotiated).

11. Conclusions

The authors have argued that three kinds of unconscious assumptions about videotelephony have created a tangle of inappropriate expectations and unsuitable terminology, with an emphasis on technical signals instead of human sensations. As a result, four kinds of skills have been undervalued or even ignored — physical issues, pictorial values, expression of meaning, and the projection of feeling.

The key question for the future is not whether videotelephony can emulate face-to-face meeting, but what it is that suitable pictures can add, in different situations, to human interaction at a distance. There is a growing list of successful telecommunications applications of video technology, that do not fit the traditional 'telephone with television' mould.

Communication by exchanging electronic pictures differs significantly from telephony and is quite unlike television, despite a shared technology. This paper has explained key differences between speech and visual communications. Some of these differences have been ignored in early kinds of videophones. The meaning and characteristics of expression and projection, in the context of videotelephony, have been discussed.

Videophones need to be made before they can be fully designed. Here, 'making' a videophone means understanding how its physical and psycho-physical features interact with the psychological behaviour and motives of its users, and how this interaction will shape the as-yet unformed social and cultural norms of videotelephony. Much 'making' is therefore yet to be done.

Exploration of user feedback and opinion now matters more than development of technology in videotelephony, because user feedback and opinion is needed to steer technical design. The authors suspect that videotelephony has a healthy and prosperous future — but to get there, developers will need to start asking the right questions and stop answering the wrong ones.

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